ONS ALOUINI

Data Science & AI Engineering Student — Seeking a PFE / Pre-Hire Internship (2027)

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PROFESSIONAL SUMMARY

Data Science & AI engineering student at ESPRIM who designs, evaluates and deploys **end-to-end machine-learning systems** — from data collection and annotation to model training, explainability, REST APIs, web interfaces and cloud deployment. Hands-on experience in **predictive modeling, NLP, computer vision and MLOps**, gained through an AI internship at Anypli, where I delivered **NEXUS AI**, a production-ready platform powered by three integrated ML models and an LLM assistant, and through projects in medical imaging, multilingual NER, intelligent recruitment and road-safety risk prediction. Rigorous about model evaluation, data quality and explainable AI, and used to Agile teamwork. Seeking a **PFE (end-of-studies) or pre-hire internship** as an AI / Machine Learning Engineer, Data Scientist, NLP or Computer Vision Engineer.

Machine Learning · Deep Learning · NLP & NER · Computer Vision · Explainable AI · MLOps · Model Evaluation · Feature Engineering · Data Annotation · LLM Integration · REST APIs & Microservices · Full-Stack Development

PROFESSIONAL EXPERIENCE

AI & Full-Stack Developer Intern — NEXUS AI

Aug 2026 (1 month)

Anypli — Monastir, Tunisia

- Built **NEXUS AI**, an AI-powered management platform for digital agencies that unifies projects, customer support and finances, with five role-based workspaces.
- Trained and served **3 ML models** as FastAPI microservices: financial health with XGBoost (**ROC-AUC 0.932**, SHAP explanations), project-effort risk with Gradient Boosting, and multilingual ticket triage (FR/EN/AR, **89% urgency accuracy**) retrained from agent feedback.
- Developed the **Laravel 12** REST API and **React 19 / TypeScript** front end: Kanban with automatic task assignment, Gantt planning, client portal, messaging, proactive risk alerts and an LLM assistant (Groq).
- Secured and shipped the platform: Sanctum tokens, TOTP two-factor authentication, per-company data isolation, 39 PHPUnit tests; deployed on Vercel, Render (Docker) and Aiven MySQL.

Technologies: Python, scikit-learn, XGBoost, SHAP, FastAPI, Laravel, React, TypeScript, MySQL, Docker · **github.com/onsalouini/nexus-ai**

Full-Stack Developer Intern

Jun – Jul 2025

ForeverMo Group — Kairouan, Tunisia

- Designed and deployed an HR management web app (employees, departments, projects, attendance) used by **20+ employees**, replacing manual Excel tracking.
- Built a RESTful **Angular + Symfony** architecture with JWT role-based access on MySQL (Doctrine ORM), an admin statistics dashboard and an integrated chatbot.
- Delivered in **4 Agile sprints** and documented the system with UML (class, sequence, use-case diagrams).

Technologies: Angular, Symfony, TypeScript, MySQL, JWT, REST API, UML · **github.com/onsalouini/projet_stage**

ACADEMIC & PERSONAL PROJECTS

MedVision AI — Medical Image Diagnosis

github.com/onsalouini/Tumeurs_Detection

- Brain-tumor MRI classifier (EfficientNetB4, transfer learning) reaching **95.71% accuracy and 0.98 ROC-AUC** with Grad-CAM explainability; custom residual CNN for breast histopathology reaching **89.15% accuracy**.
- Served through a FastAPI backend and Streamlit interface, compatible with DICOM / PACS.

Technologies: PyTorch, EfficientNetB4, Grad-CAM, FastAPI, Streamlit, DICOM

SafeRoad AI — Road-Trip Risk Prediction *(personal project, 2026)*

github.com/onsalouini/saferoad-ai

- Built an AI platform that scores the risk of a road trip and **recommends the safest route and departure time** over the next 24 hours, using real routes (OSRM), hourly weather forecasts (Open-Meteo) and segment-by-segment risk on an interactive map.
- Trained a **LightGBM** classifier on the US Accidents dataset (**7.7M records**, 94 engineered features), predicting severe accidents with **ROC-AUC 0.89** on 300K held-out records; per-prediction **SHAP** explanations.
- Audited the model end to end — target identification, calibration, stability by year and state, threshold trade-offs (precision / recall / F1) — and **detected and neutralized a data-leakage feature**, documented in a published evaluation report.
- **FastAPI** backend (JWT auth, MongoDB, user history) and **React 19** front end (Tailwind CSS, Leaflet, charts, what-if simulator).

Technologies: Python, LightGBM, SHAP, pandas, FastAPI, MongoDB, JWT, React, Leaflet, OpenStreetMap APIs

FARDI CaseForge — Multilingual NER & Anonymization

github.com/onsalouini/fardi-caseforge

- Team NLP project (FR / AR / EN): built a **training-ready NER dataset** of 864 industrial passages and 2,486 entities over 9 business labels, enriched with 4 hand-annotated public corpora (**12,300+ entities**).
- Designed the full data pipeline: synthetic generation, pre-labeling, **Label Studio** annotation with double annotation (**Cohen's kappa 0.745**), leak-free frozen splits and exports to CoNLL / BIO / **Hugging Face Datasets**.
- Trained a **CRF baseline** with ablation studies (tokenization, normalization, learning curves) and prepared **XLM-RoBERTa** fine-tuning with label masking, plus anonymized texts and 114 teaching case sheets.

Technologies: Python, NER, CRF, XLM-RoBERTa, Hugging Face, Label Studio, Cohen's kappa

ForsaTech (A.I.R) — AI Recruitment Platform github.com/BoutheinaRj/forsatech

- Team project presented at ESPRIM's **Bal des Projets 2026**: RoBERTa-based NLP extracts information from CVs, then scores and ranks candidates against job offers with compatibility reports.

Technologies: Python, RoBERTa, NLP, FastAPI, React

MLOps HeartAI — End-to-End ML Pipeline github.com/onsalouini/mlops-heartai

- Industrialized a heart-disease model with modular code, FastAPI serving, **MLflow** tracking, **Docker**, **CI/CD** (GitHub Actions, pytest) and Elasticsearch / Kibana monitoring.

Technologies: MLflow, Docker, FastAPI, GitHub Actions, pytest, Kibana

Dental Anomaly Detection — Computer Vision github.com/onsalouini/Diagnostic_Dentaire

- Full pipeline on dental X-rays (collection, annotation, augmentation, training) for a **YOLOv8** object detector reaching **88–92% mAP**.

Technologies: YOLOv8, OpenCV, PyTorch

Other Full-Stack Applications

- **Intelligent Supplier-Invoice Management Platform** — end-to-end application, defended before a jury.
- **LivraisonApp** — delivery management in ASP.NET Core MVC (C#, EF Core, Identity) with role-based access and revenue dashboard.
- **MAKOGpath** (2024–2025) — web & mobile home-healthcare booking app (Symfony, MySQL, JavaScript), delivered in Agile sprints.

EDUCATION

Engineering Degree in Computer Science — Data Science & AI Sep 2024 – 2027 (expected)
ESPRIM – École Supérieure Privée d'Ingénierie et de Technologies Monastir, Tunisia

- **Consistently ranked among the top students of my class.** Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, MLOps, Software Engineering.

Preparatory Cycle — Physics & Chemistry Sep 2022 – Jun 2024
Faculty of Sciences of Monastir (FSM) — passed the National Competitive Exam for Engineering Schools Monastir

Baccalaureate in Experimental Sciences — With Honors 2022
Lycée Ibn Rachik Kairouan, Tunisia

TECHNICAL SKILLS

Machine & Deep Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CNNs, EfficientNet, Transfer Learning, Pandas, NumPy
Model Evaluation & XAI	ROC-AUC, PR-AUC, F1, Calibration, Threshold Tuning, Data-Leakage Detection, SHAP, Grad-CAM
Computer Vision	YOLOv8, OpenCV, Object Detection, Image Classification, Medical Imaging (MRI, X-ray, DICOM)
NLP & LLMs	Transformers (RoBERTa, XLM-R), NER, CRF, Hugging Face, TF-IDF, Text Classification, Label Studio, LLM APIs (Groq)
MLOps & Cloud	MLflow, Docker, FastAPI, CI/CD (GitHub Actions), pytest, Streamlit, Kibana, Vercel, Render
Web Development	React, Angular, Tailwind CSS, TypeScript, Node.js, Laravel, Symfony, ASP.NET Core, REST APIs, JWT
Programming & Data	Python, Java, C, C#, PHP, JavaScript · MySQL, MongoDB, SQLite · Power BI
Tools & Methods	Git, Linux, UML, Agile / Scrum, Trello

CERTIFICATIONS & ACTIVITIES

- **Hedera Hashgraph (3 certifications)** — Certified Developer Associate (HCDA), Certified Foundation – Technical (HCF), Business Foundation (HBF)
- **Getting Started with AI on Jetson Nano** — NVIDIA Deep Learning Institute
- **AI Fluency: Framework & Foundations** — Anthropic
- **AI Fundamentals: Foundations for Understanding AI**
- **Insight Cybersecurity Hackathon** — participant, IEEE ESPRIM Student Branch, Nov 2025

LANGUAGES & SOFT SKILLS

Languages	Arabic (native), French (fluent, C1), English (professional, B2)
Soft skills	Teamwork, time management, problem solving, adaptability